

Learning objectives

In this lesson you will learn to:

- Differentiate between wired and wireless connectivity
- Explain how data is transmitted along copper and fibre-optic cables
- Compare the performance of copper and fibre-optic cables and give examples of their use
- Describe how high-speed broadband is delivered.

For more information and additional learner activities see Chapter 4.1 of the student book.

Transmitting data

What are some of the ways to transmit data between devices?

Think about the different ways of transmitting data between smartphones, games consoles, desktop computers, laptops, cameras, etc.

Wired vs wireless

All methods of transmitting data can be considered as either wired or wireless.

The difference is simple:

- wired transmission involves making a physical connection from one device to another, for example using a USB or ethernet cable
- wireless transmission uses no physical connection. Instead it uses radio waves, for example Bluetooth[®], Airdrop, and NFC.

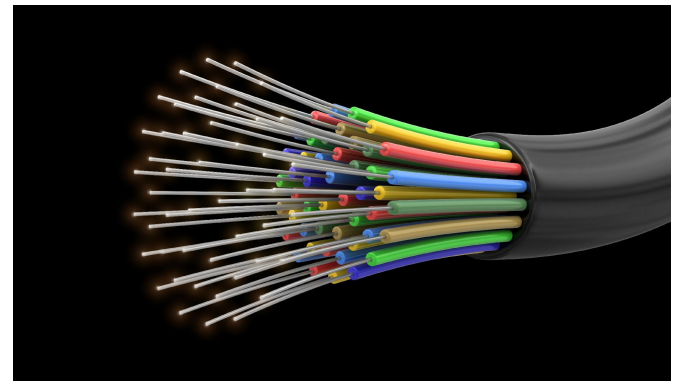
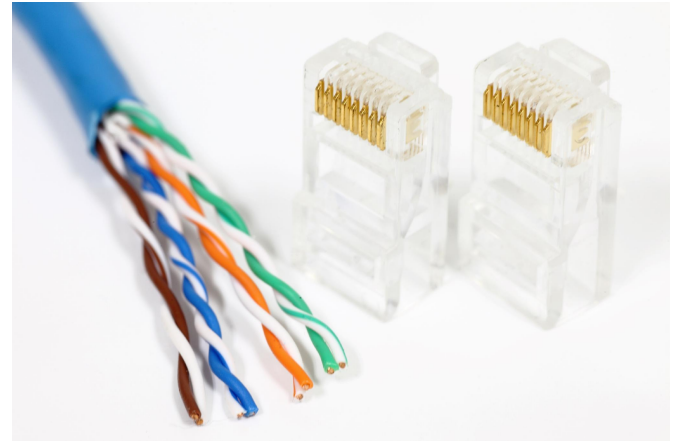
Cables

Wired transmission commonly uses copper or fibre-optic cables.

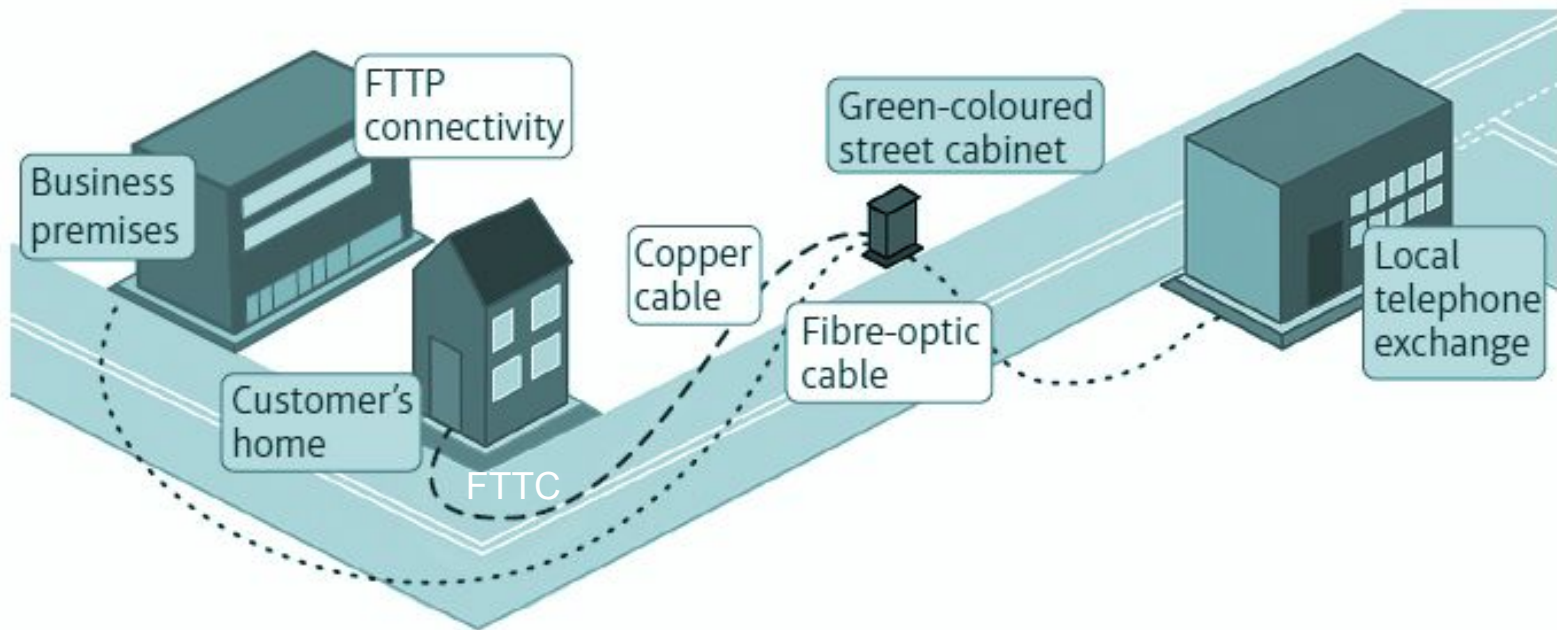
Copper is more common as it has been used to transmit data since the telegraph was invented over 150 years ago.

They can also be designed as Unshielded Twisted Pair (UTP) cables. More commonly used with connected computers together.

Fibre-optic cable is a modern alternative that has vastly higher bandwidth and lower latency.



Broadband



Broadband

Broadband is a type of high-capacity internet connectivity that is being rolled out across the country. Many of you will have broadband internet at home.

Typically for homes, broadband is implemented as **fibre to the cabinet (FTTC)**. In this approach, fibre optic cables run to green street cabinets (which you will probably have noticed). Copper cables then link homes to the cabinets (usually over distances of less than a mile).

The alternative, which is generally used for businesses, is **fibre to the premises (FTTP)**. In this approach, the final stretch from the cabinet to the business uses another fibre optic cable.

Wrap up: you have learned how to...

- ✓ Differentiate between wired and wireless connectivity.
 - The difference is in the type of connection between devices.
- ✓ Explain how data is transmitted along copper and fibre-optic cables.
 - In copper cables, data is transmitted as electrical pulses.
 - In fibre optic cables, data is transmitted as pulses of light.
- ✓ Compare the performance of copper and fibre-optic cables and give examples of their use.
- ✓ Describe how high-speed broadband is delivered.
 - Either FTTP or FTTC.